FOLDING POLYOMINOES INTO UNIFORM-THICKNESS RECTANGLES

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ABSTRACT. We consider the question of what polyominoes can be folded into a rectangle with a specified thickness, for all polyominoes up to pentominoes. We also briefly look at the thickness-2 case for hexominoes.

CONTENTS

1. Introduction

Rectangles are known to be tile-able highly symmetrical shapes, making them highly applicable for real-world scenarios. Meanwhile, thickness, or the number of layers of paper that a shape is composed of, is highly relevant in material design, since higher thicknesses can cause folding problems but preserve structural integrity, making uniform-thickness constructions — constructions for which the thickness is uniform across the shape — particularly useful. In this paper, we analyze whether each polyomino — shapes very commonly found in modular components — can be folded into uniform-thickness rectangles.

2. A note on notation

Sometimes, it is of benefit to notate polyominoes inline. For polyominoes up to pentominoes, each has a canonical letter name, and thus we can notate them by number concatenated with letter.

When dealing with polyominoes in general, we simply record each line as a sequence of `1`'s and `0`'s separated by `/`'s. Each `/` denotes a new line. Each lines starts from the same position on the left, and writes to the right, with `1` denoting filled in space and `0` denoting empty space. For instance, the `5Y` pentomino below can be denoted as `01/1111`.

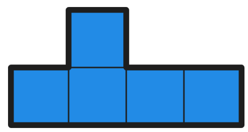

Figure 1. `5Y` pentomino, or `01/1111`

Sometimes, we may also describe a folded state. If it happens that everything is still aligned to the grid, we may simply use this notation, but rather replace the numbers with the number of layers of paper. For instance, if we fold the top cell of the `5Y` pentomino down, then we can denote the result with `1211`.

3. Monominoes

3.1. `1I` monomino. We start with monominoes, of which there are exactly one, the `1I` monomino:

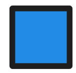

Figure 2. `1I` monomino

It is clear that we can get a rectangle of arbitrary thickness t by folding the unit square into a $1 \times 1/t$ rectangle with an operation called *crimping*. We give an example for the case where $t = 7$:

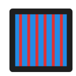

Figure 3. `1I` monomino solved with $t = 7$

Hence, the `1I` monomino is solvable for all t. (We assume positive integer t.)

We will now abbreviate all occurences of "thickness a" to "ta".

4. Dominoes

4.1. `2I` domino. Consider the only domino, `2I`:

Figure 4. `2I` domino

Recognize that the same solution for `1I` carries over, making a $1 \times 2/t$ rectangle. Here is yet again an example where $t = 7$:

Figure 5. `2I` domino solved t7

Hence, the `2I` domino is solvable for all t.

In general, any $a \times b$ rectangle can be solved tt by folding into an $a \times b/t$ rectangle of thickness t. Therefore, we know that, by reduction, if we can solve any particular shape tt, then we can solve it tkt by solving it tt, and then crimping the tt rectangle. Call this the **Reduction Lemma**.

5. Trominoes

5.1. `3I` tromino. Consider the rectangular tromino, `3I`:

Figure 6. `3I` tromino

The shape is trivially solvable t1, so solvable for all t by Reduction Lemma.

5.2. `3L` **tromino.** Consider the other tromino, `3L`:

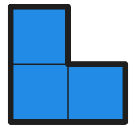

Figure 7. `3L` tromino

Clearly, since this shape is not rectangular, it cannot be solved t1. We now show that it can be solved for all other t, i.e. it is solvable for all non-degenerate t.

Now, consider the following crease:

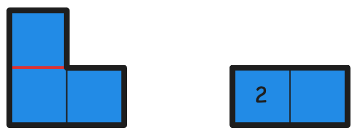

Figure 8. `3L` tromino solved t7 (part 1)

After the crease, we obtain a domino in which one cell has t2. Here, observe that this domino is solvable for all non-degenerate t. Let $t = 2a + b$, where $b < 2$. We can crimp the t2 cell, achieving a t$2a$ portion. Then, we can fold the single cell over b times, achieving a t$(2a + b)$, or tt, portion. If there are still leftover t1 portions, we can simply crimp them into tt.

Here is the case for $t = 7$:

Figure 9. `3L` tromino solved t7 (part 2)

Hence, the `3L` tromino is solvable for all non-degenerate t.

We can generalize this result by considering the following shape:

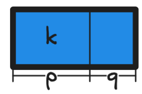

FIGURE 10. The two parts have lengths p and q, with thicknesses k and 1

Therefore, if we have some thickness $t = ak+b$, we can make the same operation, crimping the tk into tak, folding the t1 cell back and forth b times, and then crimping any leftover from t1 to tt.

This requires, however, the we have enough t1 segments to fold back and forth. The tak segment will have length p/a, and hence we require the t1 segment to have length at least bp/a. Therefore, the condition is hence $q \geq bp/a$, or $aq \geq bp$.

We can find a sufficient condition for this by realizing that $b \leq k-1$. Hence, $a \geq p(k-1)/q$ is a sufficient condition since $a \geq p(k-1)/q \geq pb/q$. Translating back to t, the sufficient condition on t would be $t \geq k \cdot \lceil p(k-1)/q \rceil$.

Therefore, with the configuration above, we can solve all cases with thickness $k \cdot \lceil p(k-1)/q \rceil$ and above. Call this bound $t_{\min}$, and call this result the **Crimping Lemma**.

6. Tetrominoes

6.1. **40 tetromino.** Consider the square tetromino, 40:

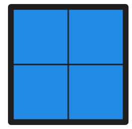

Figure 11. 40 tetromino

This is solvable t1, so by Reduction Lemma, it is solvable for all t.

6.2. `4I` **tetromino.** Consider the rectangular tetromino, `4I`:

Figure 12. `4I` tetromino

This is again solvable t1, so by Reduction Lemma, it is solvable for all t.

6.3. `4L` **tetromino.** Consider the tetromino `4L`:

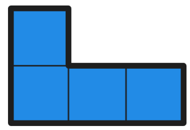

FIGURE 13. `4L` tetromino

This is clearly not solvable `t1`, but the following crease gives Crimping Lemma:

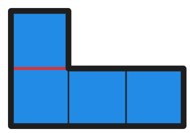

FIGURE 14. Reduction of `4L` tetromino to Crimping Lemma

with parameters

$$\begin{cases} k = 2 \\ p = 1 \\ q = 2 \\ t_{\min} = 2 \end{cases}$$

which covers all non-degenerate t.

Hence, the `4L` tetromino is solvable for all non-degenerate t.

6.4. `4T` **tetromino.** Consider the tetromino `4T`:

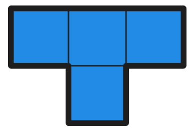

FIGURE 15. `4T` tetromino

We can row-reduce this shape, i.e. reduce this into a single row with the fold:

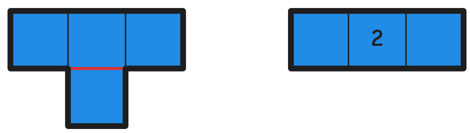

FIGURE 16. Row-reduction of `4T` tetromino

We prove that this formation can be solved for all non-degenerate t. First, observe that we can apply Crimping Lemma

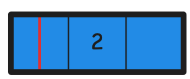

FIGURE 17. Reduction of row-reduced `4T` to Crimping Lemma

with parameters

$$\begin{cases} k = 2 \\ p = 3/2 \\ q = 1 \\ t_{\text{min}} = 4 \end{cases}$$

which covers all $t \geq 4$. Additionally, t2 and t3 are solvable with the following formations:

FIGURE 18. Row-reduced `4T` solved t2 and t3

Hence, all non-degenerate t can be solved.

6.5. `4S` **tetromino.** Observe that `4S` and `4T` share the same row-reduced form.

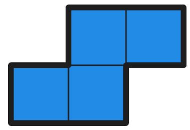

FIGURE 19. `4S` tetromino

Hence, the `4S` tetromino is also solvable for all non-degenerate t.

7. Pentominoes

7.1. `5I` pentomino.

The `5I` pentomino is as follows:

Figure 20. `5I` pentomino

`5I` is trivially solvable for all t.

7.2. `5L` **pentomino.** Consider the `5L` pentomino:

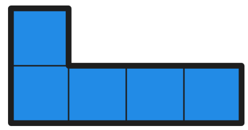

FIGURE 21. `5L` pentomino

After row-reduction, we can apply Crimping Lemma with parameters

$$\begin{cases} k = 2 \\ p = 1 \\ q = 3 \\ t_{\min} = 2 \end{cases}$$

which covers all non-degenerate t.

7.3. `5N` **pentomino.** Consider the `5N` pentomino:

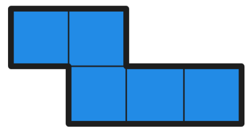

Figure 22. `5N` pentomino

After row-reduction, we can reduce to Crimping Lemma

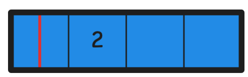

Figure 23. Reduction of row-reduced `5N` to Crimping Lemma

with parameters

$$\begin{cases} k = 2 \\ p = 3/2 \\ q = 2 \\ t_{\min} = 2 \end{cases}$$

which covers all non-degenerate t.

7.4. `5Y` **pentomino.** Consider the `5Y` pentomino:

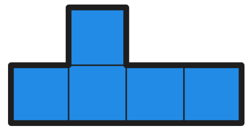

FIGURE 24. `5Y` pentomino

`5Y` has the same row-reduction as `5N`, so we apply the same logic; hence `5Y` is solvable for all non-degenerate t.

7.5. `5F` **pentomino.** Now consider the `5F` pentomino.

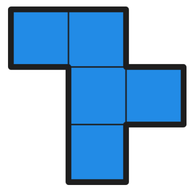

FIGURE 25. `5F` pentomino

Observe that row-reduction gives the following formation:

FIGURE 26. Row-reduction of `5F` pentomino

We now prove that this formation solves all $t \geq 3$. First, we have a reduction to Crimping Lemma

FIGURE 27. Reduction of row-reduced `5F` to Crimping Lemma

with parameters

$$\begin{cases} k = 3 \\ p = 4/3 \\ q = 1 \\ t_{\min} = 9 \end{cases}$$

covering all $t \geq 9$. We can find solutions for t3, t4, t5, and t7, for which Reduction Lemma gives t6 and t8, as follows:

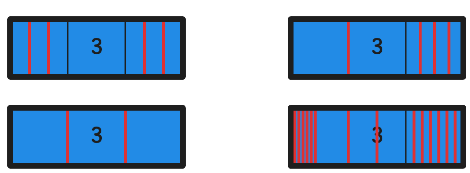

FIGURE 28. Row-reduced `5F` solved t3, t4, t5, and t7

Notably, the solution for t7 requires realizing that $7 = 2 \times 3 + 1$, folding accordingly, and finally crimping the remaining areas of t1. Hence, we have proved that the row-reduction of `5F` solves all $t \geq 3$.

Additionally, `5F` is solvable $t = 2$ as follows:

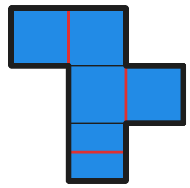

Figure 29. `5F` pentomino solved t2

Therefore, `5F` is solvable for all non-degenerate t.

7.6. `5X` pentomino. Consider the `5X` pentomino.

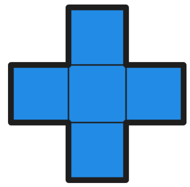

FIGURE 30. `5X` pentomino

Observe that `5X` has the same row-reduction as `5F`, which solves all $t \geq 3$. Now, let us consider the case for $t = 2$. Here is a solution of the `5X` pentomino t2:

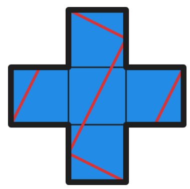

FIGURE 31. `5X` pentomino solved t2

Hence, `5X` is solvable for all non-degenerate t. However, how can one find these solutions? Well, consider the planar tiling with `5X` pentominoes:

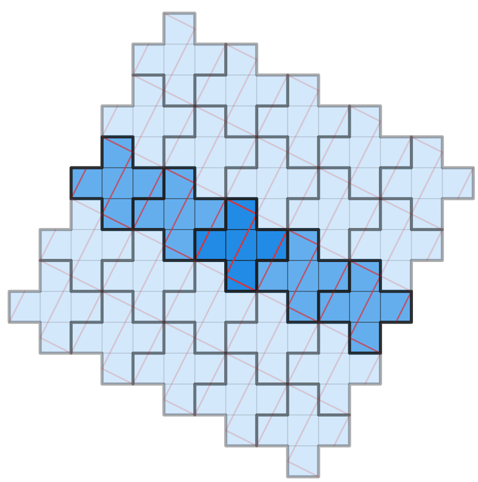

FIGURE 32. Tiling of `5X` pentomino

Notice how one strip of the planar tiling would be a long rectangle, except that the edges are jagged. Hence, we can neutralize these jagged edges by folding along the hypothetical straight edge of the long rectangle, then canceling the jagged edges with a fold perpendicular to the hypothetical straight edge, and finally completing the rectangle. Call this the **Tessellation Method**.

Omega_3301 independently discovered and formalized the Tessellation Method; upon research, one can find mention of this discovery in "Polygons Folding to Plural Incongruent Orthogonal Boxes" due to Jun Mitani and Ryuhei Uehara, when the box degenerates into a doubly-covered rectangle, a specific case of a thickness-two rectangle. (The distinction is that, after glued, doubly-covered rectangles are topologically spherical, whereas thickness-two rectangles do not necessarily have this property.) Omega_3301 provided a generalization of this method.

Upon applying the Tessellation Method again, we will formulate this method with rectangular approximants.

7.7. `5Z` pentomino. Consider the `5Z` pentomino.

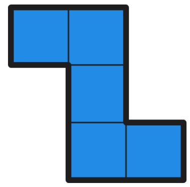

FIGURE 33. `5Z` pentomino

Observe that `5Z` again has the same row-reduction as `5F`, which solves all $t \geq 3$. Now, let us consider the case for $t = 2$, with the Tessellation Method. Below is a tessellation of `5Z`, with the strip and its rectangular approximant drawn:

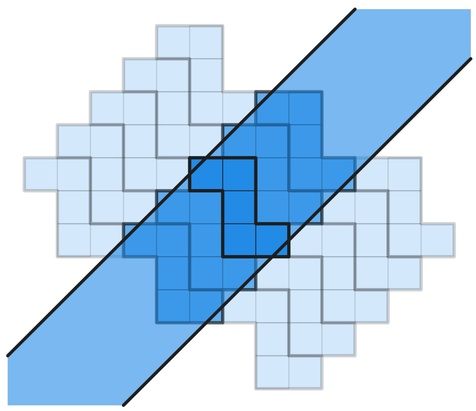

FIGURE 34. Tiling of `5Z` pentomino

Now, we can simply complete the t2 solution by drawing the creases along the rectangular approximant and neutralizing the jagged edges, creating a rectangle:

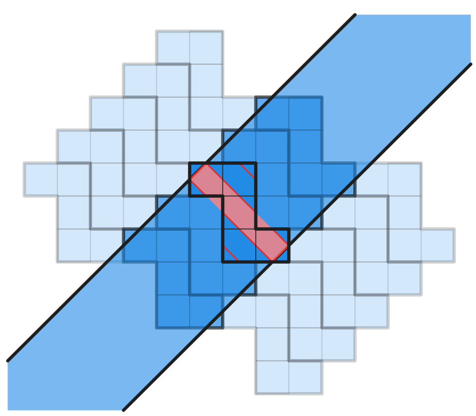

FIGURE 35. `5Z` pentomino solved t2

Observe that the created rectangle has dimensions $1/\sqrt{2} \times 5/\sqrt{2}$, and hence area $5/2$, matching our expectation (a rectangle of uniform thickness 2 folded from a shape of area 5 should have area $5/2$, so this is expected). Hence `5Z` is solvable for all non-degenerate t.

This result is generalizable. Call a *down-right path* polyomino, or just a *path* polyomino, as a polyomino produced by starting at some cell, and then repeatedly extending the polyomino by joining cells downwards or rightwards. Any path polyomino can be solved t2, with the tiling strip below:

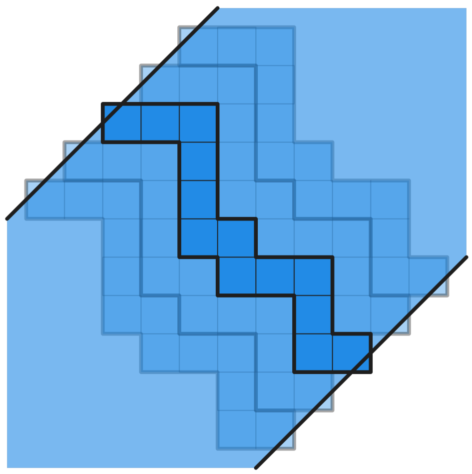

FIGURE 36. Tiling of a general path polyomino

We can thus apply the same concept of rectangular approximants, to obtain:

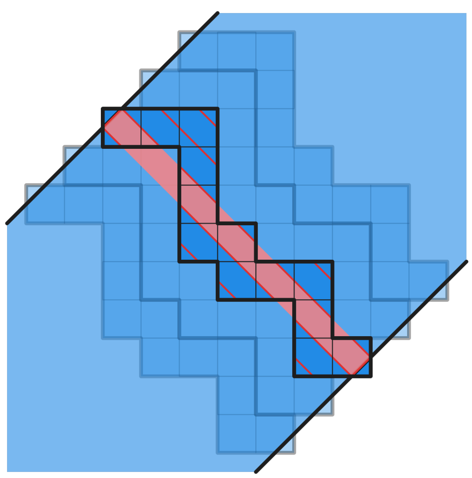

FIGURE 37. General path polyomino solved t2

The rectangular approximant may sometimes look less natural, but the same result holds regardless:

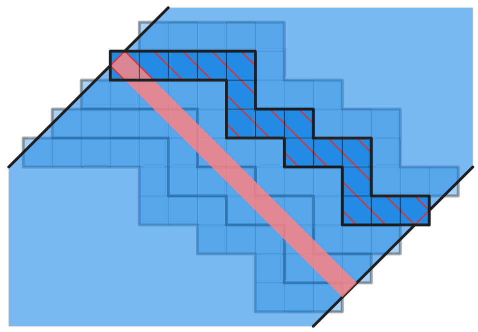

Figure 38. General path polyomino solved t2

7.8. `5V` **pentomino.** Consider the `5V` pentomino.

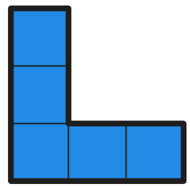

FIGURE 39. `5V` pentomino

Observe that `5V` has a row-reduction to Crimping Lemma with parameters

$$\begin{cases} k = 3 \\ p = 1 \\ q = 2 \\ t_{\min} = 3 \end{cases}$$

covering all $t \geq 3$. Additionally, since `5V` is a path pentomino, the $t = 2$ case is also solved. Hence, `5V` is solvable for all non-degenerate t.

7.9. 5T **pentomino.** Consider the 5T pentomino.

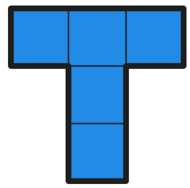

FIGURE 40. 5T pentomino

Again 5T has the same row-reduction as 5F, solving $t \geq 3$. We solve for the $t = 2$ case by a generalization of the Tessellation Method as presented by Omega_3301 which allows 180°-rotated copies in the tiling, with the idea being to split each pair of non-rotated and rotated copy after they have been folded:

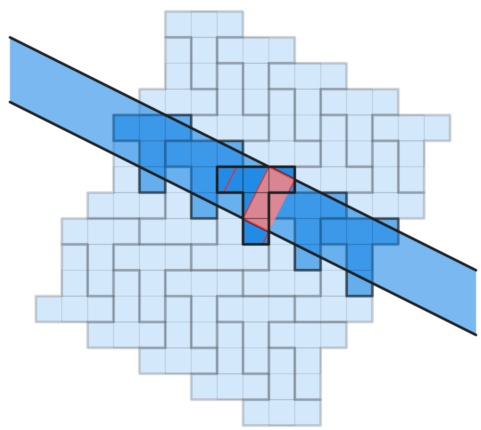

FIGURE 41. 5T pentomino solved t2

Hence, 5T is solvable for all non-degenerate t.

7.10. **`5U` pentomino.** Consider the `5U` pentomino.

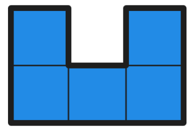

FIGURE 42. `5U` pentomino

It is clear that there is a reduction to Crimping Lemma with parameters

$$\begin{cases} k = 4 \\ p = 1 \\ q = 1 \\ t_{\min} = 12 \end{cases}$$

covering all $t \geq 12$. First, let us check $t = 3$, $t = 5$, $t = 7$, and $t = 11$:

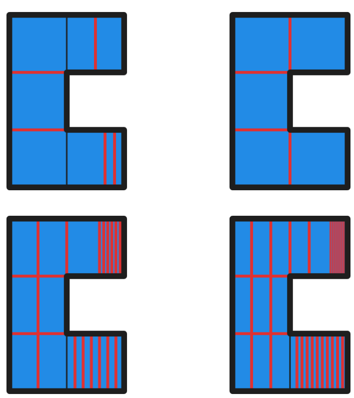

FIGURE 43. `5U` solved t3, t5, t7, and t11

It may be seen what is happening when $t = 3$ or $t = 5$, but the scenarios for $t = 7$ and $t = 11$ are hard to follow. Hence, here are the per-step instructions for $t = 7$ and $t = 11$:

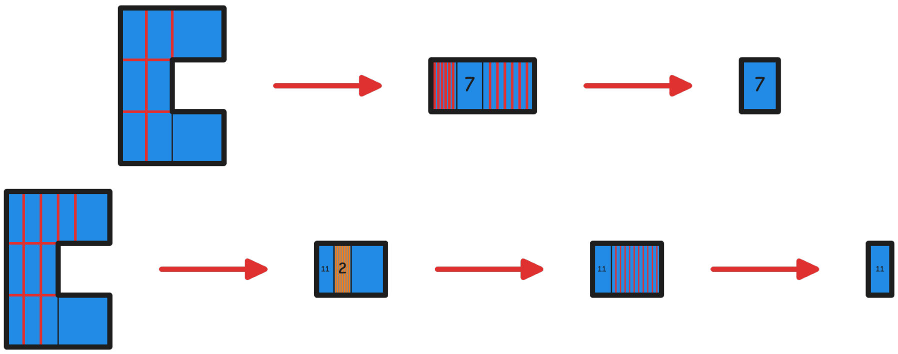

FIGURE 44. `5U` solved t7 and t11

Note the orange folds. Even though the area at which the orange folds are executed has t2, it is composed of 2 separate, essentially independent t1 sheets. These orange folds fold only one of those sheets. Hence, by crimping the section 10-fold, we get a t10 section, which, combined with the other independent t1 sheets, gives an overall t11 region.

Now, what remains to be considered is the case where $t = 2$. We apply the Tessellation Method:

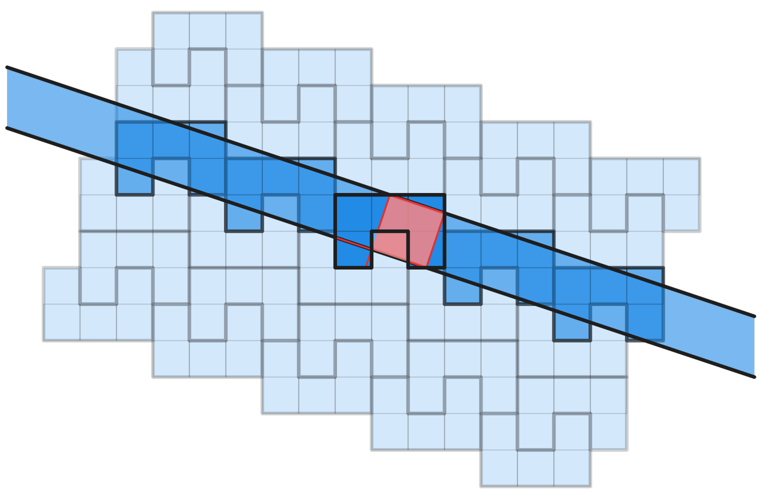

FIGURE 45. `5U` pentomino solved t2

Hence, `5U` is solvable for all non-degenerate t.

7.11. `5W` **pentomino.** Consider the `5W` pentomino.

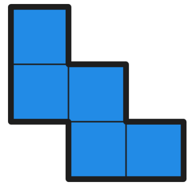

FIGURE 46. `5W` pentomino

Observe that `5W` has a row-reduction to Crimping Lemma with parameters

$$\begin{cases} k = 2 \\ p = 2 \\ q = 1 \\ t_{\min} = 4 \end{cases}$$

covering all $t \geq 4$. Additionally, since `5W` is a path pentomino, the $t = 2$ case is also solved. Hence, `5W` is solvable for all non-degenerate $t \neq 3$.

We have not found a solution, nor a disproof, of the case where $t = 3$; this case remains open.

7.12. `5P` pentomino. Consider the `5P` pentomino.

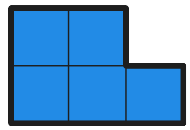

FIGURE 47. `5P` pentomino

`5P` has the same row-reduction, solving the cases where $t \geq 4$. Additionally, we can find this trivial solution to the t2 case:

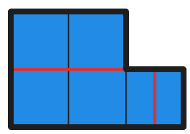

FIGURE 48. `5P` pentomino solved t2

Hence, `5P` is also solvable for all non-degenerate $t \neq 3$.

We have not found a solution, nor a disproof, of the case where $t = 3$; this case remains open.

8. A brief overview on t2 cases of hexominoes

The case for hexominoes are generally complex, but we can concentrate on certain t2 cases. Here is an overview of our results:

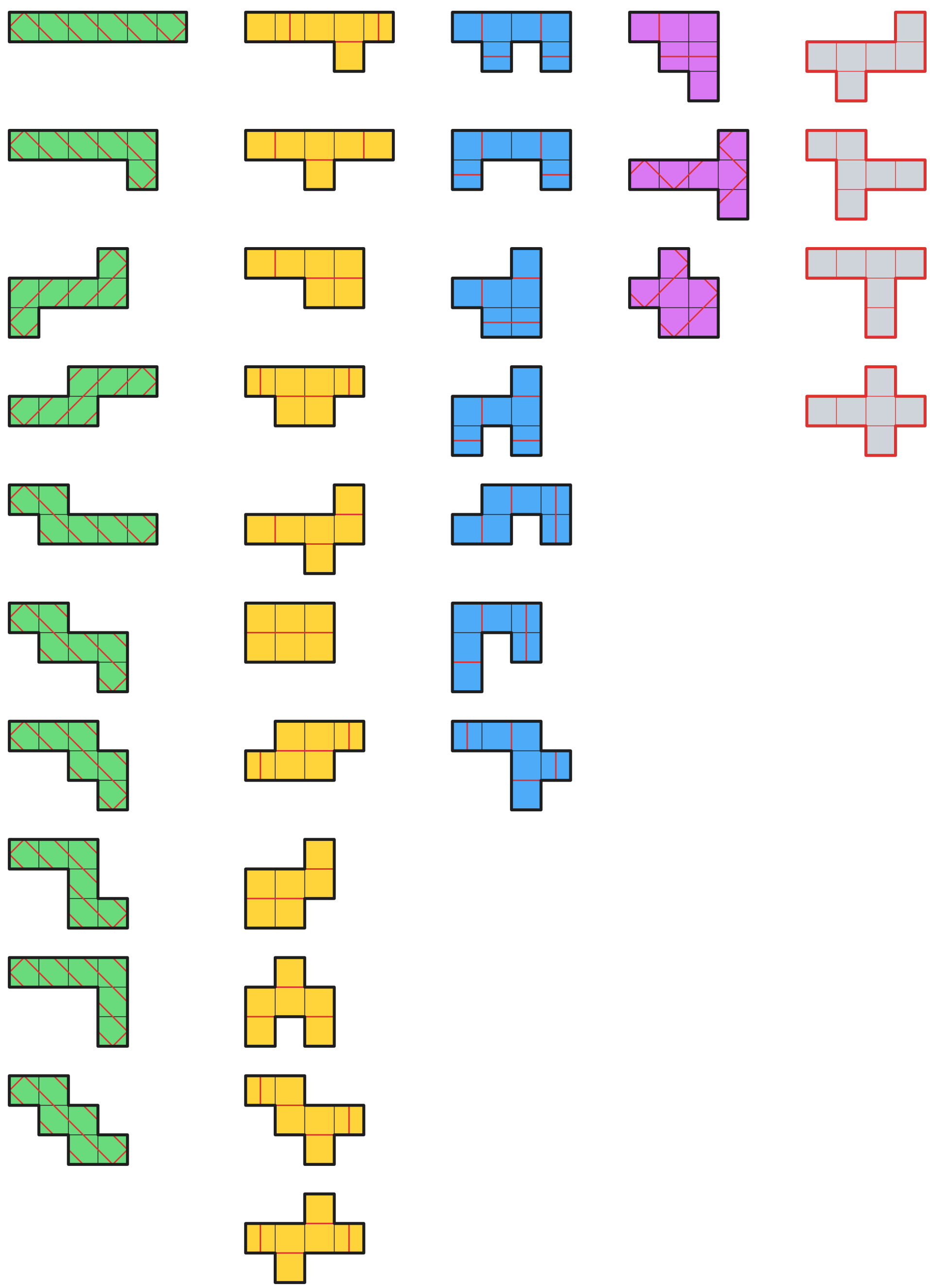

Figure 49. An overview of t2 cases in hexominoes

The first column, highlighted in green, are the path hexominoes. The second column, highlighted in yellow, are hexominoes whose row-reduction has a contiguous portion of t2 in the middle, potentially with t1 ends on either side. This row-reduction is then trivially solvable t2. The third column has hexominoes which can be reduced to the figure below, which can be solved t2.

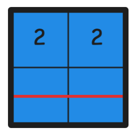

FIGURE 50. Reduction into `22/11`

The fourth column contains hexominoes which are outliers to the first three shortcuts, yet still solvable. The fifth column contains hexominoes for which we do not have solutions for, and suspect that none exist.

In particular, the latter two hexominoes of the fourth column can be solved by Tessellation Method:

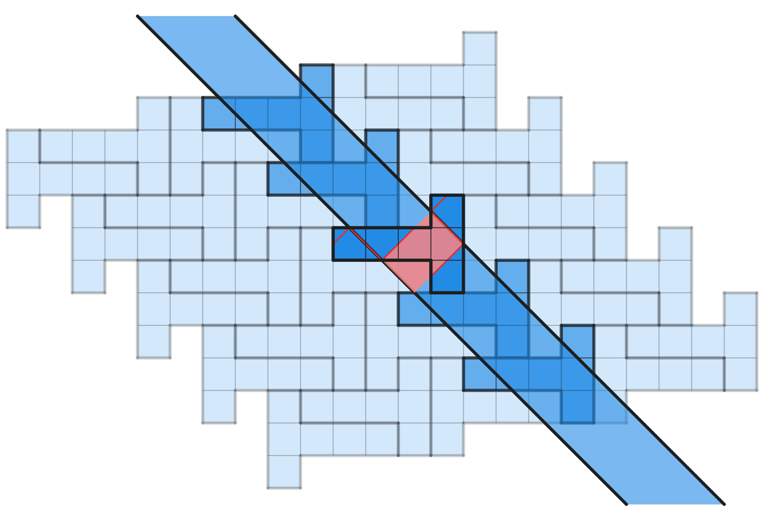

FIGURE 51. `0001/1111/0001` hexomino solved t2

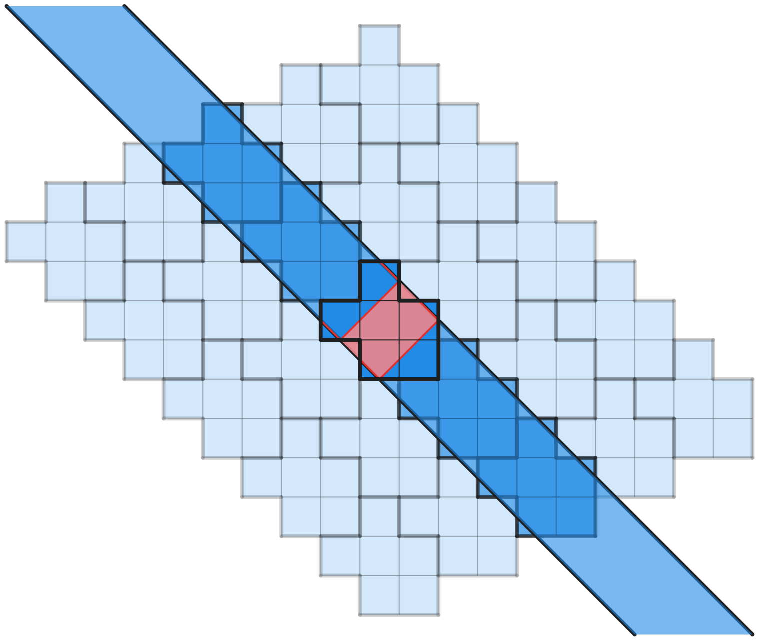

FIGURE 52. `01/111/011` hexomino solved t2

9. Conclusion

In this paper, we attempt to categorize the solvability of Origami Puzzles up to pentominoes, providing conclusive results for all but two cases, namely `5W` and `5P` pentominoes under thickness 3. Perhaps some proof exists which shows the impossibility of thickness 3, perhaps with an argument regarding 3-adic numerals. Future research may also focus on other tiles, such as triangular tiles. Furthermore, it is interesting to ask if Origami Puzzles in general is hard. The authors have a proof of NP-completeness of Origami Puzzles under only orthogonal and 45°-diagonal folds which is yet to be formalized, but is curious as to whether such results can be generalized. It may also be interesting to consider Hirata's Half-Length Theorem as applied to thickness-2 rectangles rather than doubly covered rectangles, though the applicability is questionable.

It is also natural to ask other questions, such as what specific conditions allow the Tessellation Method to succeed.

10. Acknowledgements

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All visuals were generated using Excalidraw.